

The antibacterial protection provided by the bactericidal lectin relies on its recognition of cell wall peptidoglycan (PGN), an insoluble polymer consisting of linear saccharides joined by short sequences of amino acids. The bactericidal lectin binds to specific regions in the carbohydrate portion of PGN via a nonterminal glutamate-proline-asparagine (EPN) motif in the lectin. To examine the effect of saccharide chain length on lectin binding, the scientists enzymatically cleaved insoluble PGN into smaller fragments, which were separated by size-exclusion chromatography. The cleaved, soluble fragments were then incubated with solutions containing the bactericidal lectin to assess binding. The results are shown in Table 1.

PGN saccharide length (number of sugar monomers)	K_d (mM)
5	5.28 ± 0.89
15	1.45 ± 0.31
35	0.59 ± 0.14

✓ ☒ A. glutamate's R-group pK_a plus or minus 2 pH units. [46%]

☐ B. proline's R-group pK_a plus or minus 2 pH units. [6%]

☐ C. asparagine's R-group pK_a plus or minus 2 pH units. [15%]

☐ D. the average of glutamate's amino-group pK_a and asparagine's carboxyl-group pK_a , plus or minus 2 pH units. [31%]

Explanation:



In addition to amino and carboxyl groups, some amino acids have **ionizable R groups**. As a result, the net charge of an individual amino acid or protein is determined by the pH of its environment and the pK_a values of its amino, carboxyl, and (when present) ionizable R groups.

The question asks for the pH range over which the EPN motif would go from being approximately 99% protonated to approximately 1% protonated, based on the [Henderson-Hasselbalch equation](#). **Only** the **glutamate** (Glu) residue **in** the **EPN motif** has an **ionizable R group**, and the **effect of** the pH on the **deprotonated-to-protonated ratio** is [logarithmic](#). Applying the Henderson-Hasselbalch equation to glutamate yields:

$$\text{pH} = \text{p}K_{\text{a}(\text{Glu})} + \log \left(\frac{[\text{deprotonated Glu}]}{[\text{protonated Glu}]} \right)$$

Because the logarithm of 99 (corresponding to 1% protonation) is approximately 2 and the logarithm of $\frac{1}{99}$ (corresponding to 99% protonation) is approximately -2, the **correct range** is **glutamate's R-group $\text{p}K_{\text{a}}$ plus or minus 2 pH units**.

(Choices B and C) Proline and asparagine do not have ionizable R groups, so their protonation states are not affected by pH.

(Choice D) Because of their participation in [peptide bonds](#), the amino and carboxyl groups of nonterminal amino acids in a protein (or individual protein subunits in multimeric proteins) are not ionizable and do not participate in acid-base reactions.

Educational objective:

The Henderson-Hasselbalch equation relates a chemical species $\text{p}K_{\text{a}}$ to the pH of its environment. Each unit change in a number's $\log_{10}$ corresponds to an order-of-magnitude change in the number itself.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403409
System:
Amino Acids and Proteins

Lectins are carbohydrate-binding proteins found in all life forms. Certain lectins produced by gastrointestinal tract cells participate in defense against invading pathogens. To investigate the role of these endogenous lectins in gastrointestinal physiology, a team of scientists performed experiments with a small intestinal bactericidal lectin.

The antibacterial protection provided by the bactericidal lectin relies on its recognition of cell wall peptidoglycan (PGN), an insoluble polymer consisting of linear saccharides joined by short sequences of amino acids. The bactericidal lectin binds to specific regions in the carbohydrate portion of PGN via a nonterminal glutamate-proline-asparagine (EPN) motif in the lectin. To examine the effect of saccharide chain length on lectin binding, the scientists enzymatically cleaved insoluble PGN into smaller fragments, which were separated by size-exclusion chromatography. The cleaved, soluble fragments were then incubated with solutions containing the bactericidal lectin to assess binding. The results are shown in Table 1.

Table 1 Dissociation Constant (K_d) for PGN with Saccharides of Different Lengths Bound to the Bactericidal Lectin (Note: The standard error of the mean (SEM) is given along with the K_d for each PGN species.)

PGN saccharide length (number of sugar monomers)	K_d (mM)
5	5.28 ± 0.89
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In certain bactericidal lectins that are similar to the one that was tested, carbohydrate binding is mediated by an X-P-X motif, in which X refers to amino acids that replace E and N in the EPN motif. The best explanation for proline's presence in both motifs is that:

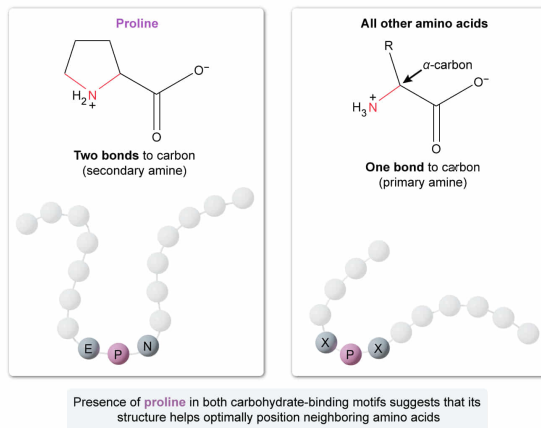
- ☐ A. proline's side chain charge facilitates ionic interactions with carbohydrates. [6%]
- ☐ B. proline's side chain provides flexibility in the positioning of its neighboring amino acids. [28%]
- ☒ C. a secondary amino group is required for favorable positioning of the neighboring amino acids. [52%]
- ☐ D. aliphatic amino acids with larger side chains are more hydrophobic, which prevents carbohydrate binding. [12%]

Correct

51% answered correctly

Explanation:

As the only secondary amino acid, proline imparts unique structural characteristics to proteins



Each **amino acid** possesses a unique R group bonded to its α -carbon. Because the other substituents of the α -carbon (ie, a hydrogen, an amino group, and a carboxyl group) are invariant between amino acids, the variable R groups give each amino acid its identity and **distinct characteristics**.

The passage describes a bactericidal lectin possessing a motif consisting of the sequence glutamate-proline-asparagine (ie, EPN), and the question asks for the best explanation for proline's presence both in the EPN motif and in similar lectins with X-P-X motifs. **Proline** is **unique** among the 20 **amino acids** commonly found in proteins in that it is the **only one that possesses a secondary amino (imino) group**. Therefore, the best explanation for proline's presence in both motifs is that a **secondary amino acid** is **necessary for optimal positioning** of its **neighboring amino acids**.

(Choice A) Proline's aliphatic side chain is *not* charged, and cannot participate in ionic interactions with carbohydrates (which also do not carry charge).

(Choice B) Because proline's side chain links to both its amino group *and* its α -carbon, its flexibility in the positioning of its neighboring amino acids is *reduced* compared to other amino acids.

(Choice D) Other aliphatic amino acids (eg, **valine**, **leucine**) are similar in size and hydrophobicity to proline. Therefore, it is more likely the unique structure of proline is required for carbohydrate binding instead of its size or hydrophobicity.

Educational objective:

As the only secondary amino acid, proline imparts structural rigidity to polypeptide regions in which it is found.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403410
System:
Amino Acids and Proteins

Lectins are carbohydrate-binding proteins found in all life forms. Certain lectins produced by gastrointestinal tract cells participate in defense against invading pathogens. To investigate the role of these endogenous lectins in gastrointestinal physiology, a team of scientists performed experiments with a small intestinal bactericidal lectin.

The antibacterial protection provided by the bactericidal lectin relies on its recognition of cell wall peptidoglycan (PGN), an insoluble polymer consisting of linear saccharides joined by short sequences of amino acids. The bactericidal lectin binds to specific regions in the carbohydrate portion of PGN via a nonterminal glutamate-proline-asparagine (EPN) motif in the lectin. To examine the effect of saccharide chain length on lectin binding, the scientists enzymatically cleaved insoluble PGN into smaller fragments, which were separated by size-exclusion chromatography. The cleaved, soluble fragments were then incubated with solutions containing the bactericidal lectin to assess binding. The results are shown in Table 1.

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PGN saccharide length (number of sugar monomers)	K_d (mM)
5	5.28 ± 0.89
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If the unique properties of each amino acid in the EPN motif are necessary for carbohydrate binding, which of the following amino acids would best substitute for the asparagine residue?

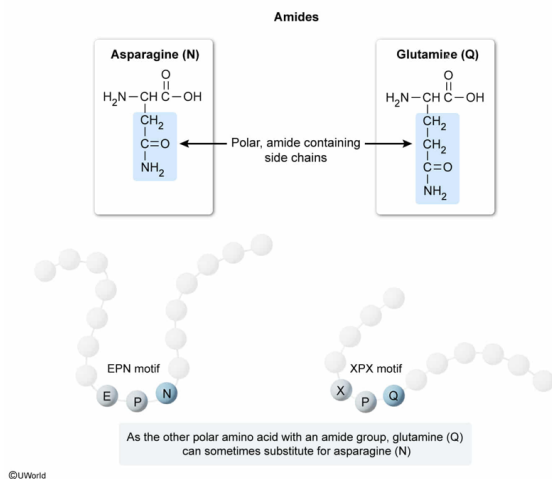
- ☐ A. Phenylalanine [1%]
- ✓ ☒ B. Glutamine [81%]
- ☐ C. Aspartate [2%]
- ☐ D. Lysine [14%]

Correct

81% answered correctly

Explanation:

Glutamine can substitute for asparagine in similar X-P-X motifs



Amino acids can be divided into **classes** based on shared properties such as polarity, charge, and chemical groups. These distinct properties are manifested through the **R groups**, and the amino acids within a particular class have R groups that are similar in one or more respects. Because of these similarities, in a certain type of mutation (called a "conservative mutation") amino acids with shared properties can sometimes **substitute** for each other in functional proteins.

The passage describes an EPN motif within a bactericidal lectin, and the question asks for the amino acid that would best replace the asparagine if the unique properties of each amino acid in the EPN motif are necessary for carbohydrate binding. Among the 20 most common amino acids in proteins, **asparagine is one of the two with polar, amide-containing R groups. The other such amino acid is glutamine**; therefore, **glutamine would best replace asparagine in a similar motif**.

(Choice A) Because **phenylalanine** is a nonpolar, aromatic amino acid, it is not a good replacement for a polar, amide-containing amino acid like asparagine.

(Choices C and D) **Aspartate** is the negatively charged ionic form of the acidic amino acid aspartic acid, and **lysine** is a basic amino acid whose side chain is positively charged at physiological pH. Therefore, neither aspartate nor lysine would be a good replacement for asparagine, whose R group is uncharged.

Educational objective:

Of the 20 most common amino acids found in proteins, glutamine and asparagine are the only two that are polar and contain amide side chains.

Time spent: 0 sec

Subject:

Biochemistry

QID: 403411

System:

Amino Acids and Proteins

Lectins are carbohydrate-binding proteins found in all life forms. Certain lectins produced by gastrointestinal tract cells participate in defense against invading pathogens. To investigate the role of these endogenous lectins in gastrointestinal physiology, a team of scientists performed experiments with a small intestinal bactericidal lectin.

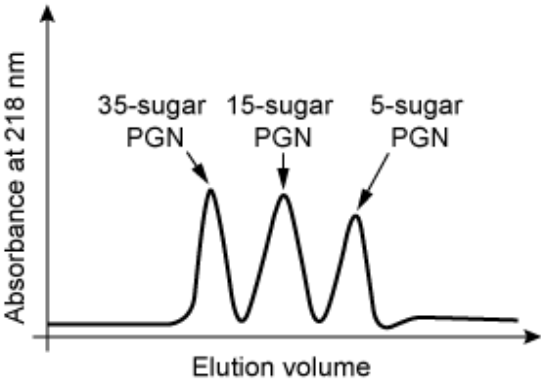
The antibacterial protection provided by the bactericidal lectin relies on its recognition of cell wall peptidoglycan (PGN), an insoluble polymer consisting of linear saccharides joined by short sequences of amino acids. The bactericidal lectin binds to specific regions in the carbohydrate portion of PGN via a nonterminal glutamate-proline-asparagine (EPN) motif in the lectin. To examine the effect of saccharide chain length on lectin binding, the scientists enzymatically cleaved insoluble PGN into smaller fragments, which were separated by size-exclusion chromatography. The cleaved, soluble fragments were then incubated with solutions containing the bactericidal lectin to assess binding. The results are shown in Table 1.

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5	5.28 ± 0.89
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35	0.59 ± 0.14

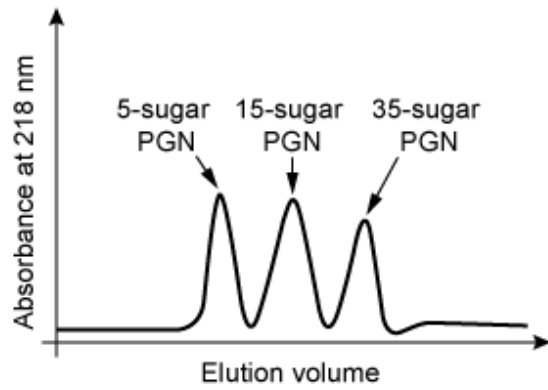
Carbohydrates absorb ultraviolet light at 218 nm. Which of the following figures depicts the expected sequence of peaks detected by ultraviolet absorption as the cleaved PGN fragments elute from the size-exclusion column?

☒ A.



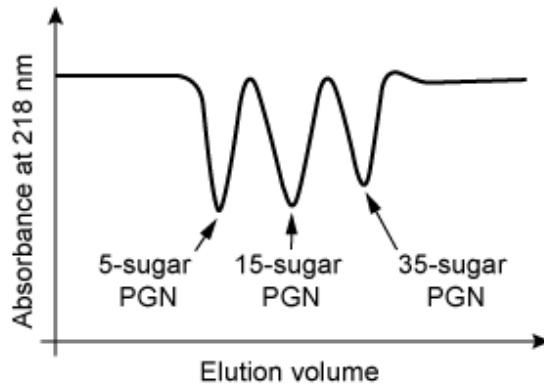
[54%]

☐ B.



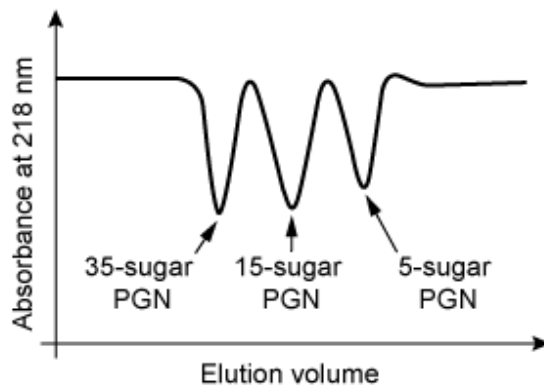
[29%]

☐ C.



[7%]

☐ D.



[9%]

Correct

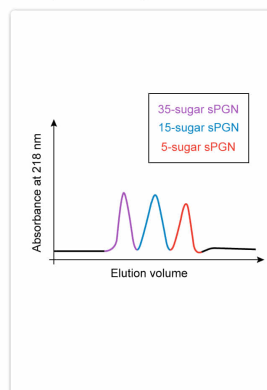
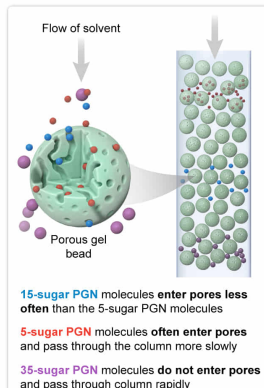
53% answered correctly

Explanation:

The 35-sugar PGN fragments elute first & the 5-sugar PGN fragments last

① Separation by size-exclusion chromatography

② Detection using UV absorbance



Common structural components of molecules **absorb light** differently at different wavelengths, and this property can be used to help identify the structural components of biological molecules. This absorbance is often used to identify molecules in a mixture after they are separated based on different properties (eg, charge, size).

Size-exclusion chromatography is a technique that **separates** molecules **according to size**. This separation is accomplished by applying the sample to a special column with porous beads. As the sample passes through the column, **smaller molecules enter the pores** and their **progress is slowed** whereas **larger molecules cannot enter the pores** and thus **pass through** the column **more quickly**.

The passage describes an experiment in which PGN fragments with saccharide chains of various lengths were separated using size-exclusion chromatography, and the question asks for the graph that correctly depicts the absorbance peaks of the PGN fragments exiting the column. PGN molecules with the **largest saccharide chains** pass through the column quickest (corresponding to **less total elution volume** collected), and PGN molecules with the **smallest saccharide chains** pass through the column the slowest (requiring a **greater elution volume**). Therefore, the graph shown in Choice A best depicts the expected results.

(Choice B) Because the time required for passage of a molecule through a size-exclusion column is inversely proportional to the size of the molecular species being separated, the elution volume collected is *smallest* for the 35-sugar PGN and *greatest* for the 5-sugar PGN.

(Choices C and D) The question states that carbohydrates absorb 218 nm light. These graphs show absorbance *decreasing* as each molecule elutes (ie, more light is transmitted), but they should show absorbance *increasing* (ie, peaks should appear rather than troughs). In addition, the order of the eluting peaks in Choice C is the reverse of the expected order.

Educational objective:

Size-exclusion chromatography separates molecules according to size, with elution order inversely related to size (ie, larger molecules elute faster and small molecules elute slower).

Time spent: 0 sec
Subject:
Biochemistry

QID: 403412
System:
Amino Acids and Proteins

Lectins are carbohydrate-binding proteins found in all life forms. Certain lectins produced by gastrointestinal tract cells participate in defense against invading pathogens. To investigate the role of these endogenous lectins in gastrointestinal physiology, a team of scientists performed experiments with a small intestinal bactericidal lectin.

The antibacterial protection provided by the bactericidal lectin relies on its recognition of cell wall peptidoglycan (PGN), an insoluble polymer consisting of linear saccharides joined by short sequences of amino acids. The bactericidal lectin binds to specific regions in the carbohydrate portion of PGN via a nonterminal glutamate-proline-asparagine (EPN) motif in the lectin. To examine the effect of saccharide chain length on lectin binding, the scientists enzymatically cleaved insoluble PGN into smaller fragments, which were separated by size-exclusion chromatography. The cleaved, soluble fragments were then incubated with solutions containing the bactericidal lectin to assess binding. The results are shown in Table 1.

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PGN saccharide length (number of sugar monomers)	K_d (mM)
5	5.28 ± 0.89
15	1.45 ± 0.31
35	0.59 ± 0.14

The researchers hypothesized that the bactericidal lectin recognizes the intact PGN of live bacteria better than soluble PGN fragments from dead bacteria *in vivo*. Which of the following findings would be most consistent with this hypothesis?

- ✔ ☒ A. The apparent affinity of the lectin for full-length saccharides is greater than for any single binding site within the saccharide. [69%]
- ☐ B. Cooperative binding to soluble PGN fragments with single lectin binding sites increases the apparent affinity of the lectin for each site. [15%]
- ☐ C. Binding the lectin to a soluble PGN saccharide with a single lectin binding site increases dissociation of the lectin from full-length PGN. [13%]
- ☐ D. The binding affinity of the lectin for soluble or insoluble PGN is the same when the total number of lectin binding sites are the same. [1%]

Correct

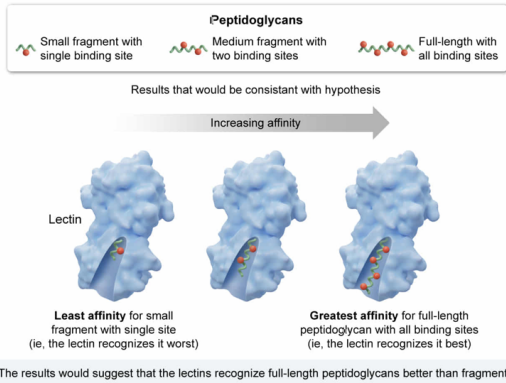
68% answered correctly

Explanation:

Testing a hypothesis about the recognition of peptidoglycan by the bactericidal lectin

Hypothesis: The bactericidal lectin recognizes full-length intact bacterial peptidoglycan better than peptidoglycan fragments

Experiment: Measure affinity of bactericidal lectin for peptidoglycans of varying lengths



Recognition of invading pathogens is an essential aspect of immune defense. The passage describes a bactericidal (ie, bacteria-killing) lectin produced by intestinal cells. This lectin recognizes bacterial cell wall peptidoglycan, a polymer of linear saccharides joined by short amino acid sequences, and binds to it.

The question states that researchers hypothesized that the bactericidal lectin recognizes the intact PGN of live bacteria better than soluble PGN fragments from dead bacteria *in vivo*, and it asks for the experimental findings most consistent with this hypothesis. Because the lectin recognizes peptidoglycan through binding, **greater binding affinity** would be **consistent with better recognition**. Therefore, **finding that the lectin binds the full-length peptidoglycan with greater affinity than it binds to peptidoglycan fragments** would be **consistent with** the researchers' hypothesis.

(Choice B) Demonstration of **cooperative binding** to soluble PGN fragments with single lectin binding sites would not address whether the lectin recognizes intact PGN better than soluble PGN fragments. This result would only show that binding to one PGN fragment improves binding to other PGN fragments.

(Choice C) Increased dissociation of the lectin from full-length PGN due to prior binding to a saccharide with a single lectin binding site would indicate *decreased* affinity. This result would indicate *reduced* recognition of intact PGN; therefore, it would not support the researchers' hypothesis.

(Choice D) Finding that the affinity of the lectin is the same for soluble or intact PGN under these conditions provides no basis for preferential recognition of intact PGN by the lectin. Therefore, this finding would not provide support for the researchers' hypothesis.

Educational objective:

Comparison of binding affinities is an appropriate approach to testing the hypothesis that a protein recognizes certain ligands better than others.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403413

System:
Amino Acids and Proteins

Lectins are carbohydrate-binding proteins found in all life forms. Certain lectins produced by gastrointestinal tract cells participate in defense against invading pathogens. To investigate the role of these endogenous lectins in gastrointestinal physiology, a team of scientists performed experiments with a small intestinal bactericidal lectin.

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5	5.28 ± 0.89
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Which interpretation(s) is(are) consistent with the data in Table 1?

- I. The affinity of PGN with a 5-sugar saccharide is significantly less than that of PGN with a 35-sugar saccharide.
 - II. The affinity of PGN with a 15-sugar saccharide is significantly greater than that of PGN with a 35-sugar saccharide.
 - III. The affinity of PGN with a 15-sugar saccharide is significantly less than that of PGN with a 5-sugar saccharide.
- ☒ A. I only [78%]
☐ B. II only [2%]
☐ C. III only [3%]
☐ D. II and III only [15%]

Explanation:

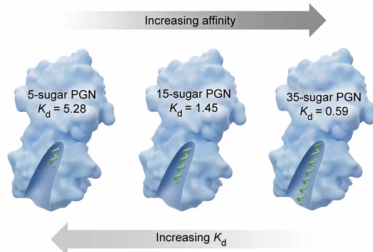
The affinity of PGN with a 5-sugar saccharide is significantly less than that of PGN with a 35-sugar saccharide

PGN saccharide length (number of sugar monomers)	K_d (mM)	$K_d - 2\text{SEM}$	$K_d + 2\text{SEM}$
5	5.28 ± 0.89	3.50	7.06
15	1.45 ± 0.31	0.83	2.07
35	0.59 ± 0.14	0.31	0.87

Range defined by $K_d \pm 2\text{SEM}$ does not overlap with that for other saccharide lengths

Range defined by $K_d \pm 2\text{SEM}$ overlaps

The $K_d \pm 2\text{SEM}$ of the 5-sugar saccharide is higher than, and does not overlap with, that of the 35-sugar saccharide, indicating significantly lower affinity.



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Protein-ligand binding interactions occur between the protein's binding sites for the ligand and the epitopes on the ligand (ie, the ligand regions where binding with the protein occurs). When the structures are highly complementary, the ligand binding site on the protein is easily accessible to the ligand and allows for strong interaction between ligand and protein (sometimes **facilitated by shape changes** within the protein).

Binding affinity can be assessed by determining the **dissociation constant** (K_d) for the interaction. The K_d is a measure of the dissociation of the protein-ligand complex into separate (unbound) molecules. The value of K_d represents the ligand concentration required for 50% of the protein molecules to be bound to a ligand. Therefore, a **lower K_d** reflects **higher affinity**, and a **higher K_d** reflects **lower affinity**.

The question asks for the interpretation most consistent with the data in table 1. A **general rule** for assessing statistical differences when p -values are not given is that **groups are likely different if the mean $\pm 2\text{ SEM}$ ranges do not overlap**. Statement I, stating that the affinity of PGN with a 5-sugar saccharide is significantly less than that of PGN with a 35-sugar saccharide, is therefore correct.

(Statement II) The mean $\pm 2\text{ SEM}$ ranges of PGN with a 15-sugar saccharide and PGN with a 35-sugar saccharide overlap, indicating they are *not* statistically different using the general rule described above. In addition, the affinity of PGN with a 15-sugar saccharide is on average *less* than that of PGN with a 35-sugar saccharide based on their respective K_d values, even if the difference *were* significant.

(Statement III) The mean $\pm 2\text{ SEM}$ ranges of PGN with a 5-sugar saccharide and PGN with a 15-sugar saccharide do not overlap, indicating they are statistically different using the general rule described above. However, the affinity of PGN with a 15-sugar saccharide is on average *greater* than that of PGN with a 5-sugar saccharide, based on their respective K_d values.

Educational objective:

When assessing binding interactions, a lower K_d reflects higher affinity and a higher K_d reflects lower affinity. As a general rule when exact p -values are not given, groups are statistically different if their mean ± 2 SEM ranges do not overlap.

Time spent:0 sec

Subject:
Biochemistry

QID:403414

System:
Amino Acids and Proteins